

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester III Sessional Examination, 2025 (NEP)

PHYSICS (SEC 3)**Paper Code: PHYSE31 : Computational Physics — Python Programming (Sessional)**

Time: 55 Minutes

Maximum Marks: 20

*Instructions: Use the Python programming language to address the following problems.***Question 1.** Print the following outputs: [1+1+1]

- a) your name
- b) your roll number
- c) A finite sequence X with elements $a_n = n^2$ where $n = [1, 10]$

Question 2. Find the mean of the finite sequence $X = \{n^2\}$ where $n = [1, 10]$. [3]*Hint:* $(a_n)_{\text{mean}} = \mu = \frac{a_1 + a_2 + a_3 + \dots + a_N}{N}$ **Question 3.** Find the standard deviations of the finite sequence $X = \{n^2\}$ where $n = [1, 10]$. [4]*Hint:* $(a_n)_{\text{std}} = \sigma \equiv \sqrt{\frac{(a_1 - \mu)^2 + (a_2 - \mu)^2 + (a_3 - \mu)^2 + \dots + (a_N - \mu)^2}{N}}$ **Question 4.** What is the use of negative indexes? Demonstrate it using an example of your choice? [2]**Question 5.** Write a python program to find the factorial of 6. [3]**Question 6.** Mention True or False for the following statement: [5]

- a) Python is written in c language.
- b) A hello function can be defined as `"def Hello():"`.
- c) A library can be imported using `"import numpy as np1"`.
- d) A numpy array can be created using `"nums = [1,2,3,4,5]"`.
- e) The output of the `"print(4 + '3')"` is 7.
